

Novel Design Concepts of Efficient Mg-Ion Electrolytes toward High-Performance Magnesium–Selenium and Magnesium–Sulfur Batteries

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Developing high-voltage Mg-compatible electrolytes (>3.0 V vs Mg) still remains to be the biggest R&D challenge in the area of nonaqueous rechargeable Mg batteries. Here, the key design concepts toward exploring new boron-based Mg salts in a specific way of highlighting the implications of anions are proposed for the first time. The well-defined boron-centered anion-based magnesium electrolyte (BCM electrolyte) is successfully presented by facile one-step mixing of tris(2H-hexafluoroisopropyl) borate and MgF_2 in 1,2-dimethoxyethane, in which the structures of anions have been thoroughly investigated via mass spectrometry accompanied by NMR and Raman spectra. The first all-round practical BCM electrolyte fulfills all requirements of easy synthesis, high ionic conductivity, wide potential window (3.5 V vs Mg), compatibility with electrophilic sulfur, and simultaneously noncorrosivity to coin cell assemblies. When utilizing the BCM electrolyte, the fast-kinetics selenium/carbon (Se/C) cathode achieves the best rate capability and the sulfur/carbon (S/C) cathode exhibits an impressive prolonged cycle life than previously published reports. The BCM electrolyte offers the most promising avenue to eliminate the major roadblocks on the way to high-voltage Mg batteries and the design concepts can shed light on future exploration directions toward high-voltage Mg-compatible electrolytes.

However, the research of rechargeable Mg batteries still remains at a nascent stage and there is a lack of high-performance Mg-compatible electrolytes and high energy density cathodes, which severely impede the research progresses in this field.

Designing an efficient Mg electrolyte is not trivial as seen from the outlined periodic table in Figure 1, where the anions in Mg salts are regimented according to common rules as follows: a high-valence central element (in most cases B, Al, P, S, Cl, As) with its bonded functional groups (such as $-\text{F}$, $-\text{Cl}$, $-\text{OR}$, $=\text{O}$, and so on) forms the well-defined anions. For instance, when Cl, P, and As elements are served as high valence centers, the as-determined anions (generally ClO_4^- , PF_6^- , and AsF_6^- , respectively) could react with Mg anode to passivate the Mg surface by generating an impervious layer (i.e., they are not compatible with Mg anode).^[4,5] Similar signature is also observed when S serves as central element (the determined anions are typically CF_3SO_3^- , TFSI^-).^[5,6]

There exist two large families of Mg-compatible electrolytes. Al-centered anion category electrolytes are termed as magnesium organohaloaluminates electrolytes, which represents the pioneering works of nonaqueous rechargeable Mg batteries developed by Aurbach et al. and Gregory et al.^[3,5] A fully inorganic Mg-salt electrolyte derived from reaction between MgCl_2 and AlCl_3 .^[7,8]

1. Introduction

Rechargeable Mg batteries offer great potential to outperform the state-of-the-art lithium-ion batteries due to their relatively large volumetric energy densities (3833 mA h cm^{-3} for Mg vs 850 mA h cm^{-3} for the commercial graphite anode).^[1–3]

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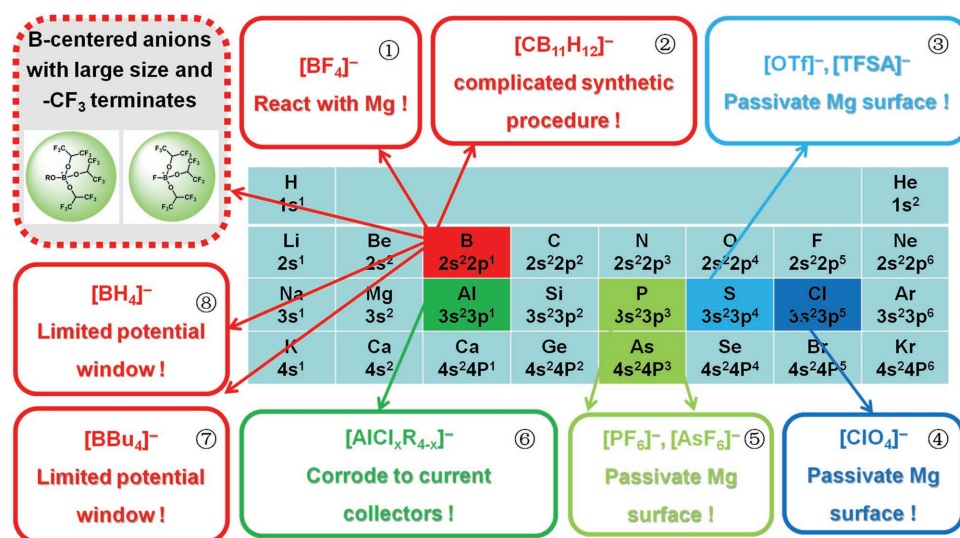


Figure 1. Decorated periodic table for directing efficient Mg-ion electrolytes. Electrolytes derived from anions of ① to ⑧ suffer from specific problems. Electrolytes from ①, ③, ④, and ⑤ could react with Mg metals forming impervious surface layers. Anions of ⑦ and ⑧ display insufficient anodic stability resulting in limited potential windows of the determined Mg-ion electrolytes. Electrolytes containing anions of ⑥ corrode to typical current collectors and show incompatibility toward sulfur and oxide cathodes. The complicated synthetic procedure and incompatibility to sulfur cathodes of ②-constructed Mg-ion electrolyte makes it less promising for practical Mg-ion electrolytes.

is also included in this large family. Boron has been proven to be another prolific element in generating efficient anions for Mg-ion electrolytes. Electrolytes based on three B-centered anions, i.e., BH_4^- , $[\text{BBu}_4]^-$, and $[\text{CB}_{11}\text{H}_{12}]^-$, are completely compatible with Mg metal.^[4,9,10] However, these all discussed Mg-compatible electrolytes encounter specific problems in regard to practical applications as depicted in colored boxes in Figure 1.

Although extensive efforts have been made and several breakthroughs in exploring novel Mg-ion electrolytes have been gained, the improvements in electrochemical potential window, compatibility with sulfur cathodes, and noncorrosivity to coin cell assemblies are still highly needed. In most cases, the anions with their solvated structures are the key determinants of electrochemical properties of Mg-ion electrolytes because they set the cell operating voltage limits, impact the charge transfer, as well as govern desirable properties of compatibility with Mg anode.^[11–14] However, little information is available in previous reports on rational design strategies for future Mg-ion electrolytes, particularly in a way of tailoring the anions of Mg salts at the molecular level.

We here for the first time propose key design concepts toward exploiting efficient anions for simple-type, high-voltage, and non-nucleophilic Mg-compatible electrolyte. In this work we focus on B-centered anions, one key aspect in which is their multifarious structural configurations, that is, it can yield large numbers of $[\text{BX}_n\text{Y}_{4-n}]^-$ ($\text{X}, \text{Y} = \text{R}, \text{RO}-, -\text{F}, \text{ and } -\text{H}$) anions via tuning both bridging and terminal groups in $-\text{X}$ and $-\text{Y}$ complexes.^[15,16] By virtue of this benefit, the premeditated design strategies are aimed at engineering B-centered anions with high compatibility with Mg anode, as well characterized in non-nucleophilic, chloride-free, and easy preparation features. Second, in order to access high soluble Mg salts in etheral solvents, monovalent large, and bulky organoborate anions should be considered in our design matrix because anions with large size facilitate salts dissociation due to their smaller lattice

energy between anion and cations.^[17,18] Finally, the electrochemical window of the Mg-ion electrolytes could be widened by grafting functional groups with high oxidation stability. Complete fluorination of terminals in $-\text{X}$ and/or $-\text{Y}$ complexes would be preferential because powerful electron-withdrawing fluorine substituents can positively affect the thermal and electrochemical stability by tuning the B-centered anions' highest occupied molecular orbital (HOMO) energies.^[19–22]

Directed by the laid out design strategies, we present the first all-round practical Mg-ion electrolyte meeting all requirements of easy synthesis, high ionic conductivity, wide operating potential window up to 3.5 V versus Mg, compatibility with electrophilic sulfur, and simultaneously noncorrosivity to typical coin cell assemblies. In the following sections, we show how our efforts have been directed to achieve this promising Mg-ion electrolyte and systematically compare the electrochemical performances with previously reported Mg-compatible electrolytes. At last, we examine the feasibility of the as-designed boron-centered anion-based magnesium (BCM) electrolyte in promising magnesium–selenium and magnesium–sulfur battery systems.

2. Results

2.1. Research Efforts in Exploring Efficient B-Centered Anions for Mg-Ion Electrolytes

The electrochemical performances of the bis[salicylato(2–)] borate magnesium/1,2 dimethoxyethane (DME) (MgBSB/DME), difluoro(oxalato)borate magnesium/DME (MgDFOB/DME), chlorotriphenylborate magnesium/DME (MgCTPB/DME), and BCM electrolytes have been investigated via cyclic voltammetric (CV) tests. MgBSB (as depicted in the inset of Figure 2a, gray color) was initially attempted for Mg-ion electrolyte (synthetic

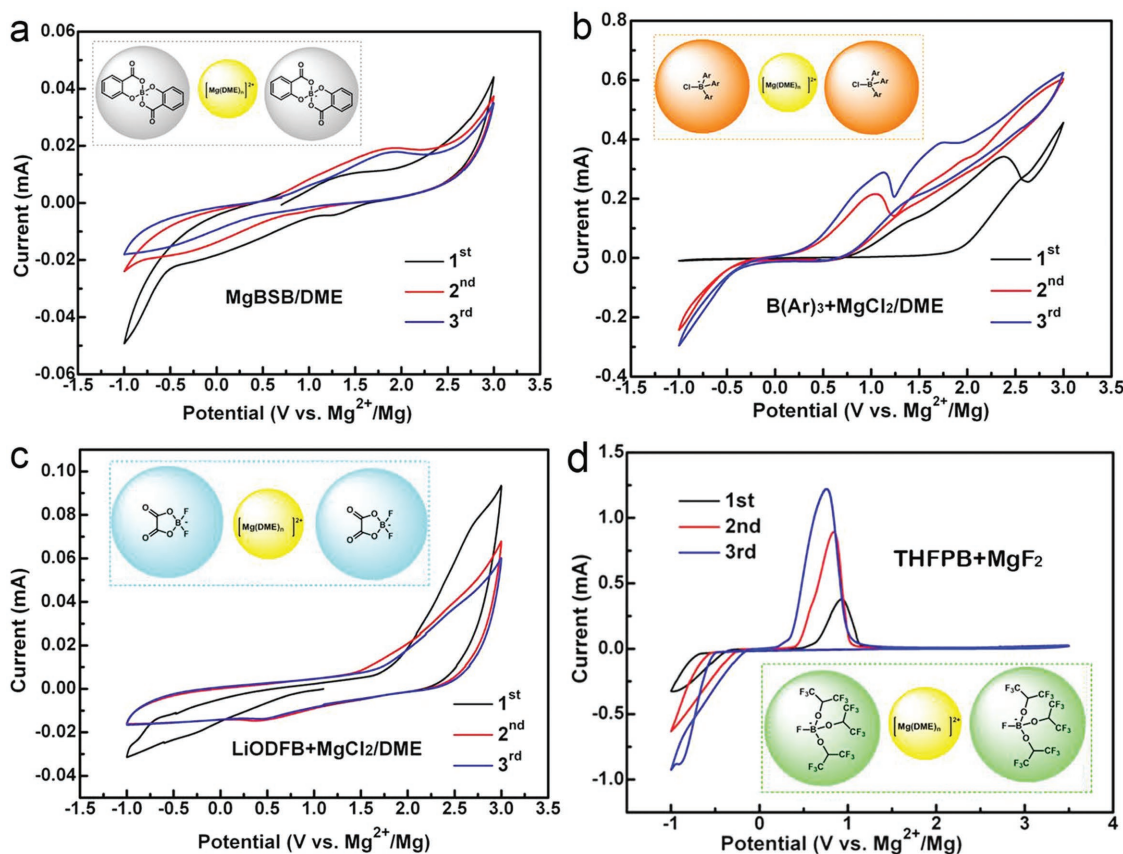


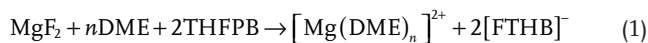
Figure 2. CV curves of SS electrode in a) MgBSB/DME electrolytes, b) MgDFOB/DME electrolyte, c) MgCTPB/DME electrolyte, and d) BCM electrolyte containing THFPB and MgF_2 at 5 mV s^{-1} . The insets display four proposed Mg salts based on B-centered anions for Mg-ion electrolytes. The gray, light blue, orange, and pale green colored anions are $[\text{BSB}]^-$, $[\text{DFOB}]^-$, chlorotriphenylborate $[\text{CTPB}]^-$, and $[\text{FTHB}]^-$, respectively. The cations are delocalized Mg^{2+} coordinated with DME solvents depicted in yellow color.

procedures; see Figure S1 in the Supporting Information and the Experimental Section). Due to the enlarged radius and pronounced charge delocalization of the $[\text{BSB}]^-$ anion, the as-derived Mg salt has fair dissolution ability in DME solutions up to 0.3 m . The reduction current becomes notable from 1.5 V in the initial CV cycle for MgBSB/DME electrolytes tested at 20°C , which indicates that the perishable anion of $[\text{BSB}]^-$ can be reduced at the anode side. Elevated temperature would be beneficial for Mg stripping/plating kinetics (Figure S2, Supporting information). However, there is almost invisible current response of the stainless steel (SS) electrode in MgBSB/DME electrolytes. The reasonable explanation might be attributed to the fact that cathodic decomposition of $[\text{BSB}]^-$ anion blocks the Mg anode by generating impervious surface layers.^[19,23] In virtue of the benefits of fluorine terminals already discussed, we directed reaction between another fluorinated borate (LiDFOB) and MgCl_2 to obtain fluorinated Mg salt based on another B-centered anion, i.e., $\text{Mg}(\text{DFOB})_2$ (as depicted in the inset of Figure 2c, light blue color). Unfortunately, the $\text{Mg}(\text{DFOB})_2$ was obtained as precipitates in DME solutions. Thus, there are no reversible Mg stripping/plating processes in CV curves of this electrolyte (Figure 2c).

A more delocalized triphenylborate ($\text{B}(\text{Ar})_3$) was directed to react with MgCl_2 to form more stable MgCTPB/DME electrolyte (inset of Figure 2b, orange color). Due to the strong

donor–acceptor effects between $\text{B}(\text{Ar})_3$ Lewis acid and Cl^- anions, a less electric field attraction between Mg^{2+} cations and Cl^- anions could be achieved leading to a large MgCl_2 salt dissociation in DME solution, leading to high concentration of MgCTPB (beyond 0.5 m) in DME. Although this electrolyte can successfully enable highly reversible Mg stripping/plating processes (Figure 2b), it is not compatible with non-noble current collectors (oxidative stability is less than 2.0 V versus Mg in typical coin cell) as a result of containing chlorides.^[3,24]

To access viable Mg-ion electrolytes, we further extend our work to another borate of tris(2H-hexafluoroisopropyl) borate (THFPB). This B-center borate is composed of electrochemically stable alkoxy bridging and $-\text{CF}_3$ terminating groups forming a large and bulky anion. Considering the chloride-free design concept, here we chose MgF_2 as Mg^{2+} cation sources in incipient species. Fortunately, partial of MgF_2 salts can be effectively dissociated with the assistance of THFPB borate since the affinity between center element of boron in THFPB and F^- anions are strong enough to break up the ionic bonds in MgF_2 crystal as depicted in equation as below (where n is quantified with scales of 3–6 in consideration of the overall Mg^{2+} coordination number to ether oxygens possibly to be five or six.)



The formed anions are shown in inset of Figure 2d. Although the solubility of MgF_2 is low ($\approx 0.05 \text{ M}$ in DME), the as-determined BCM electrolyte containing THFPB and MgF_2 can be capable of reversibly electrodepositing Mg (Figure 2d) and give the highest anodic/cathodic current response compared with that of before-mentioned electrolytes. Next, the structures of the anions in BCM electrolyte as well as its electrochemical performances have been thoroughly investigated.

2.2. Compatibility with Mg Anode

Electrochemical compatibility with Mg metal anode is mandatory for any practical Mg-ion electrolyte. Typical CV tests are used to evaluate this vital property. It is clear that DME supported BCM electrolyte can be capable of reversibly electrodepositing Mg (Figure 3a) compared to other solvents including tetrahydrofuran (THF), dimethylsulfoxide (DMSO), and so on

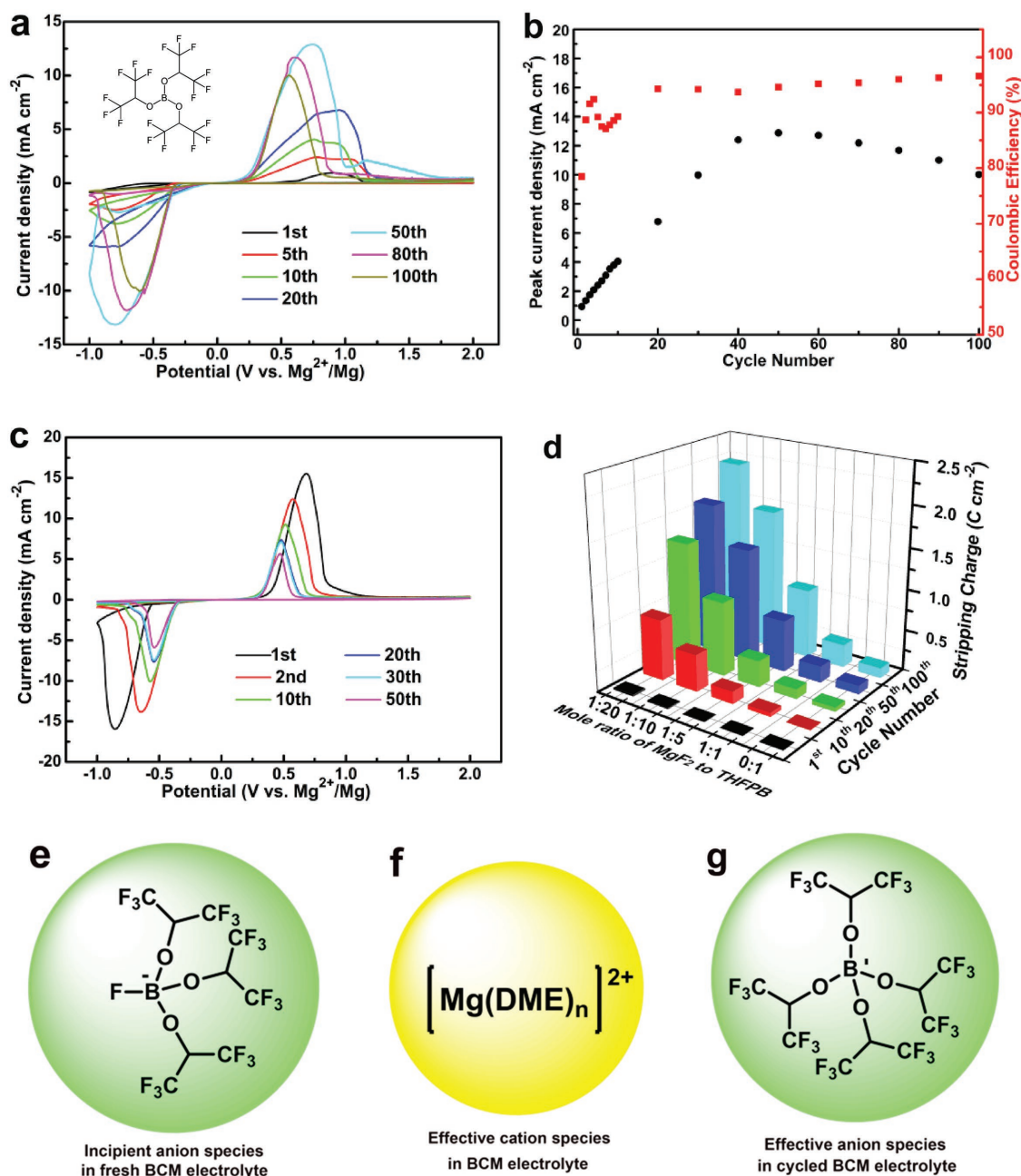
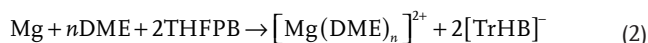


Figure 3. The electrochemical performances of the BCM electrolyte and the corresponding efficient species. a) CV curves of Mg stripping/plating behaviors in the BCM electrolyte containing $0.05 \text{ M MgF}_2 + 0.5 \text{ M THFPB}$ in DME at 5 mV s^{-1} ; inset shows the chemical structure of THFPB. b) The peak current density during Mg stripping process and the corresponding Coulombic efficiencies obtained from CV curves. c) CV curves of the BCM electrolyte $0.05 \text{ M MgF}_2 + 0.5 \text{ M THFPB}$ in DME at 5 mV s^{-1} after resting for 20 d. d) The stripping charges in the specific electrolyte systems with different mole ratios of THFPB to MgF_2 . e) The incipient anion species in fresh BCM electrolyte, f) the efficient cation species in BCM electrolyte, and g) the efficient anion species in cycled BCM electrolyte.

(see Figure S3 in the Supporting Information). The initial CV cycle (Figure 3a) displays reasonable Coulombic efficiency of 79.9% due in part to the low solubility of MgF_2 salts in DME solvents. Our mass spectrometry (MS) analysis confirms that the incipient anion species is fluoro-tris(2H-hexafluoroisopropyl)borate anion-([FTHB][−]) (as depicted in Figure 3e). During the consecutive CV cycles, the BCM electrolyte exhibits improved electrochemical properties. The Coulombic efficiency (CE) increases to a steady CE of ≈95% after 20th cycle as shown in Figure 3b. The overpotentials for Mg stripping/plating process are reduced from the second cycle onward. In addition, the peak current density examined in the 40th anodic process approaches to 12.6 mA cm^{-2} from the initial cycle of 0.89 mA cm^{-2} (Figure 3b), implicating the significant increase of active Mg species after consecutive CV cycles. Our MS results (Figure S4, Supporting Information) complemented by NMR (Figure S5, Supporting Information) and Raman (Figure S6 and Table S1, Supporting Information) investigations offer strong proofs that efficient anion and cation species in the cycled BCM electrolyte are tetra(hexafluoroisopropyl)borate anion-([TrHB][−]) (Figure 3g) and $[\text{Mg}(\text{DME})_n]^{2+}$ (Figure 3f), respectively. The formation mechanism might follow conceivable reaction below



where n is in scales of 3–6. The concentration of Mg^{2+} cation chelating with DME solvent is largely enhanced from $2.9\text{--}30.6 \text{ g L}^{-1}$ during consecutive Mg stripping/plating cycles as revealed by inductively coupled plasma (ICP) analysis. It is worth noting that the continuous increase in the peak current density up to the 40th cycle can be explained as an electrolytic conditioning process. During this process, the concentration of Mg^{2+} cation is largely improved up to 10 times of the value in fresh BCM electrolyte. Considering the limited solubility of ether-based solvents, the electrolyte solution might be saturated with the increase of extra Mg^{2+} cations, which can induce the salts precipitation on electrodes.^[5,25] This might be an important factor contributing to the moderate loss in stripping peak current density from the 50th cycle to the 100th cycle (Figure 3b).

The electrochemical performances of the BCM electrolyte do not change dramatically after 20 d resting as revealed in sequent CV curves (Figure 3c), although the peak current density slightly decreases to a steady value of 8.7 mA cm^{-2} after 10 cycles. The CEs of the aged BCM electrolyte during the 1st and 50th cycles are recorded to be 100% and 99.9%, respectively. The steady Coulombic efficiencies combined with the constant peak current density of the aged solvents indicate the high stability of active Mg species in BCM electrolyte.

The electrolyte concentrations based on stripping charges and corresponding Coulombic efficiencies have been optimized and

demonstrated in Figure 3d and Figure S7a (Supporting Information). These data unequivocally show that the stripping charges increase with the increase of THFPB concentrations, which is up to its maximum at $\text{MgF}_2/\text{THFPB}$ ratio of 1:20. A practical Coulombic efficiency of 99.8% and reasonable ionic conductivity of 1.1 mS cm^{-1} can be obtained in BCM electrolyte with $\text{MgF}_2/\text{THFPB}$ ratio of 1:20 during the 50th Mg stripping/plating cycle (Figure S7a, Supporting Information). The above results indicate that the BCM electrolyte perfectly meet the pre-requisite criteria in terms of compatibility with Mg anode.

2.3. Anodic Stability

As a reliable alternative to common Mg-ion electrolytes, wide electrochemical potential windows as well as noncorrosivity toward typical coin cell assemblies (such as the casing, springs, and current collectors) are also highly desired. Initially, we investigate the feasibility of Li-based electrolyte containing LiF salt and THFPB in DME solution, the CV curve of which (Figure S8, Supporting Information) delivers extremely high anodic stability up to 5.3 V versus Li^+/Li tested in typical coin cells. Based on the rule of thumb that puts the Mg stripping/plating at roughly 1 V above that of Li,^[26,27] the anodic stability limit for BCM electrolyte is estimated to ≈4.3 V versus Mg, far beyond that of previously reported Mg-compatible electrolytes (summarized in Table 1).^[4]

Practically, the BCM electrolyte shows an extremely high oxidative stability limit up to 3.5 V versus Mg on SS electrode as observed in Figure 4a,b when tested in coin cells, although it displays undesirable anodic stability on Cu (≈2.1 V vs Mg) and Ni electrode (≈2.2 V vs Mg). It displays improved Coulombic efficiency and higher current density in the 2nd and 3rd CV cycles seen from Figure 4b, consistent with results in Section 2.2. Linear sweep voltammetry (LSV) analyses show that commercially available Al current collector displays the highest anodic stability of 3.8 V versus Mg (Figure 4b). The widened electrochemical potential window of the BCM

Table 1. A systematic comparison of paramount parameters with previously reported Mg-compatible electrolytes.

Electrolytes	Potential window [V]	Limited potential in coin cells	Ionic conductivity at RT [mS cm^{-1}]	Coulombic efficiency [%]	Compatibility with sulfur?
APC ^{a)} [26]	3.3	2.2	4–5	100	No
$\text{Mg}(\text{BH}_4)_2$ ^[9]	1.7	1.7	2.6	100	No
MACC ^[8]	3.4	1.8	≈2.0	100	No data
HMDS–Mg ^{b)} [14]	2.5	2.4	≈2.0	>95	Yes
HMDS–Mg ^{c)} [28]	3.9	Unavailable	Unavailable	Unavailable	Yes
$\text{Mg}(\text{CB}_{11}\text{H}_{12})_2$ ^[4]	3.8	3.5	2.9	>99	No data
$\text{Mg}(\text{PF}_6)_2$ ^[29]	4.0	1.5	28.3	Unavailable	No data
BCM	3.8	3.5	1.1	99.8	Yes

^{a)}It represents APC-type electrolytes made from the transmetalation reaction between phenylmagnesium-chloride (PhMgCl) and AlCl_3 in THF; ^{b)}It represents a non-nucleophilic electrolyte formed from the reaction between hexamethyldisilazide magnesium chloride (HMDSMgCl) and aluminum trichloride (AlCl_3); ^{c)}It represents a non-nucleophilic Mg–HMDS-based electrolyte by reacting bis-hexamethyldisilazide magnesium $[(\text{HMDS})_2\text{Mg}]$ with AlCl_3 .

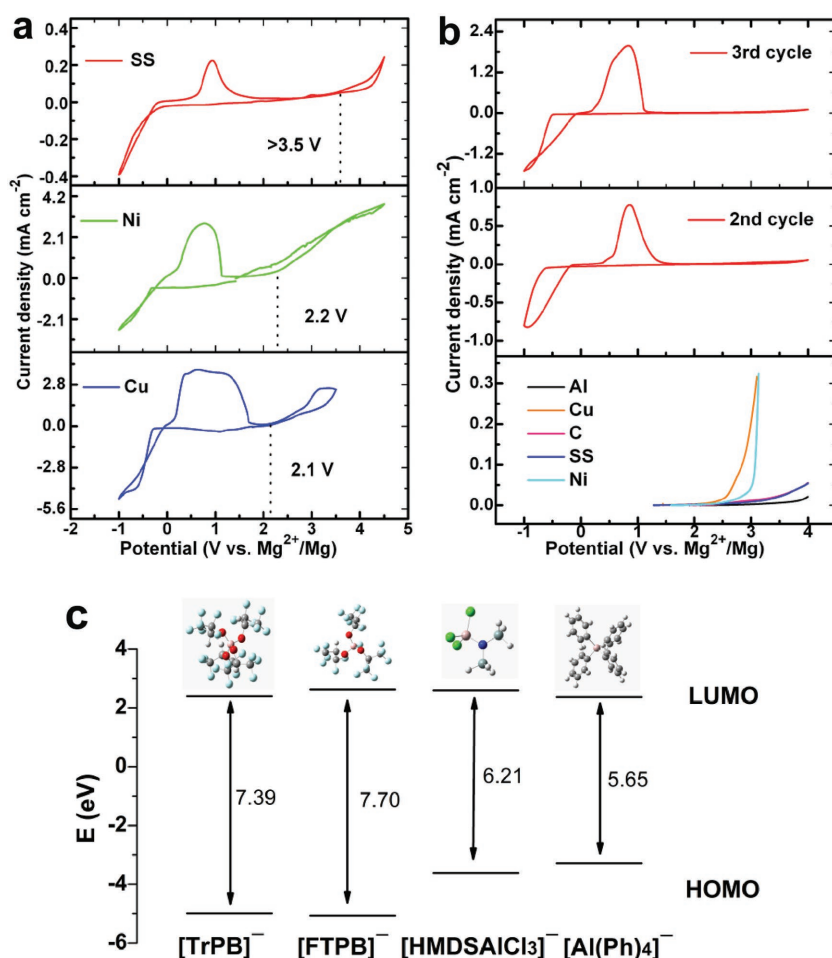


Figure 4. The oxidative stability of the BCM electrolyte. a) CV curves of BCM electrolyte at 5 mV s⁻¹ tested with SS (first cycle, red, up), Ni (green, middle), and Cu (blue, down) working electrode in typical coin cells. b) The second (up) and third (middle) Mg stripping/plating CV cycles in BCM electrolyte at 5 mV s⁻¹ within potential range from -1.0 to 4.0 V versus Mg and the LSV curves of various electrodes (downside). c) The calculated HOMO and lowest unoccupied molecular orbital (LUMO) energy levels of four well-defined anions; insets depict the four anions' geometry structures (from left to right: [TrPB]⁻, [FTPB]⁻, [HMDSAICl₃]⁻, and [Al(Ph)₄]⁻).

electrolyte could be attributed to the lowering HOMO energy levels of the fluorinated boron-centered anions compared with former electrolytes based on [hexamethyldisilazide (HMDS) AlCl₃]⁻ and [AlPh₄]⁻ as seen in Figure 4c. It is anticipated that the standard coin-type cells can be successfully constructed and charged up to 3.5 V versus Mg based on cheap SS, C, and Al current collectors, which is of vital importance to explore high-voltage cathodes for future high energy density Mg batteries. There is no pitting corrosion on SS electrodes (Figure S9a, Supporting information) and no changes in the composition of SS electrode surface (Energy dispersive X-ray spectrometry (EDX) analysis and the inset table in Figure S9b, Supporting Information). These experimental evidences combined with calculations authenticate the rationality and validity of our design concepts that complete fluorination of terminals (such as -CF₃) would be beneficial to the thermodynamic and electrochemical stability of B-centered anion-based electrolytes.

2.4. Characterizations of Mg Deposits

Dendrite-free surface is also demanded for any metal anode in a battery. X-ray diffraction (XRD) result (Figure 5a) confirms that the agglomerated deposits are Mg polycrystallines in nature. Typical scanning electron microscope (SEM) image in Figure 5b shows the galvanostatical electrodeposition products from the BCM electrolyte. It is clear that Mg deposits represent a smooth and dendrite-free morphology composed of substantial spherical Mg deposits structures in agreement with previous reports.^[4,9,30] The dendrite-free morphology (magnified SEM image in Figure 5c) poses less safety concerns compared to the hitherto unresolved Li and Na metal anode technology. EDX spectrum (Figure 5d) detects strongest Mg peak along with a spot of C, O, and F humps on Mg deposits surface in agreement with the XRD results. In addition, these analyses further confirm that the B-centered anions and DME coordinated Mg²⁺ cations are electrochemically stable against cathodic reduction leading to the absence of imperative surface layers.

2.5. Comparison with Previous Mg-Compatible Electrolytes

Here, we systematically summarize and compare the data in our work with that of currently available Mg-compatible electrolytes in Table 1 (accompanying with Figures S10 and S11, Supporting Information). It is clear that the BCM electrolyte is the only electrolyte solution that not only can be operated in typical coin cells up to 3.5 V versus Mg but also shows excellent compatibility with sulfur cathode. Considering its

facile synthesis (using commercially available materials and one-step mixing procedure), the BCM electrolyte offers great promise as the best model electrolyte for future high energy density Mg batteries.

2.6. Electrochemical Properties of Magnesium–Selenium Batteries and Magnesium–Sulfur Batteries

Here, we couple Mg anode with the Se/C and S/C cathodes (synthetic procedures; see the Supporting Information) in order to investigate the feasibility of BCM electrolyte. The Se/C composites deliver a relative low specific capacity of 422 mA h g⁻¹ at the initial cycle (Figure 6a) at 0.066 A g⁻¹. The capacity gradually increases from the 2nd cycle, which behaves as that of conversion sulfur cathodes.^[31] A reasonable explanation for the increasing capacity has been proved to be the slow penetration of electrolyte into mesoporous carbon

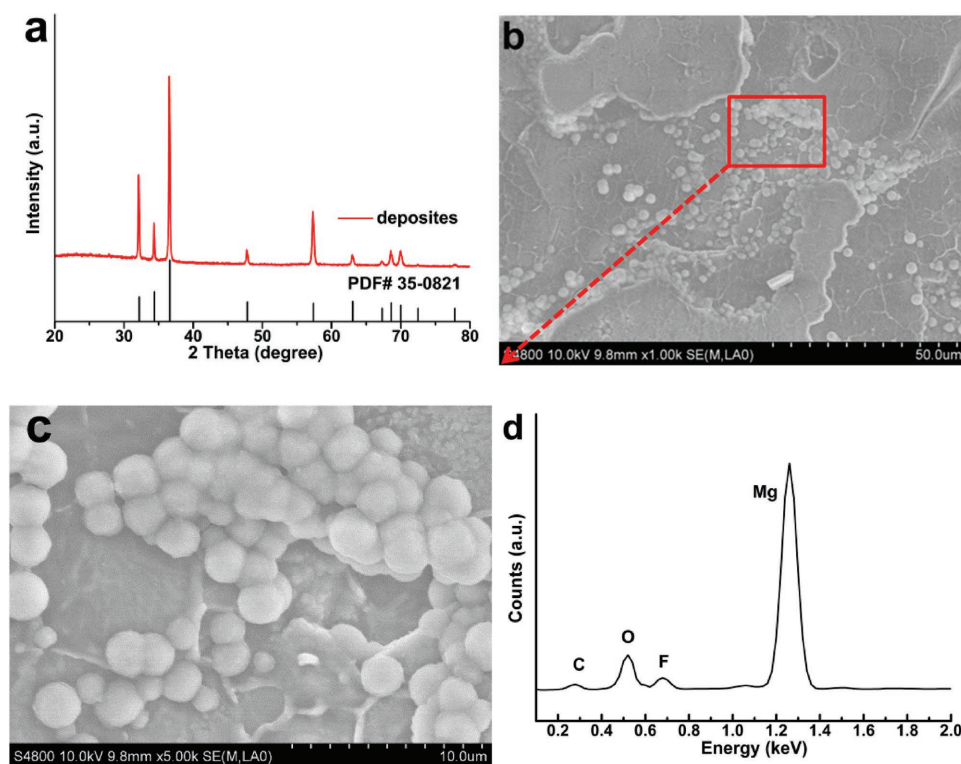


Figure 5. Characterizations of the Mg deposits. a) XRD results of the deposits. b) Typical SEM images of the deposits. c) The magnified figure in the selected area of (b). d) EDX analysis of the deposits.

substrates. In our work, another aspect has been supposed to be the main reason, i.e., the insufficient Mg depositing/stripping Coulombic efficiency of BCM electrolyte in initial cycles as presented in Figure 3. After the electrolyte conditioning, the obtained high Coulombic efficiency combined with gradually increased Se utilization yields a maximum specific capacity of 615 mA h g^{-1} at the 7th cycle. The Se/C electrode displays a stable capacity of 600 mA h g^{-1} during the consecutive cycles. The overpotential in the first cycle is up to 620 mV, which decreases to 330 mV in the 10th cycle. Figure 6b shows that the Se/C electrode displays a high average capacity of 530 mA h g^{-1} at relatively low current density of 0.05 A g^{-1} , which decreases to 512, 477, 432, and 405 mA h g^{-1} at 0.1, 0.25, 0.5, and 1.0 A g^{-1} , respectively. In this regard, the Se/C electrodes represent the best rate capability than previous reports,^[1,5,27,32] which can be truly comparable with the well-developed Li and Na batteries. Preliminary results indicate that copper in current collector plays an important role in activating the sluggish reaction between Se/C cathode and Mg anode (a detailed description of these studies is beyond the scope of this work). In addition, Se/C cathode reaches 75% capacity retention at the 100th cycle, and delivers a capacity of 361 mA h g^{-1} after 200 cycles (Figure 6c). These excellent performances exceed most reports of conversion-type cathodes for Mg batteries.^[14,31,33,28,34,35]

The magnesium–sulfur couple is also a very promising system due to its high theoretical energy density of $\approx 4000 \text{ W h L}^{-1}$, which is much higher than that of typical Li- and Na-ion batteries. The initial discharge capacity of

S/C cathode is up to 1081 mA h g^{-1} . It displays a flat voltage plateau of $\approx 1.1 \text{ V}$ versus Mg (Figure 6d) and no overcharge is observed in the following cycles. To the best of the obtainable published literatures (summarized in Table 2), the S/C cathode with the highest sulfur loading of 85% displays the best capacity retention (86.4% of its initial capacity is retained after 30 cycles in Figure 6e) as well as the slightest overpotentials (less than 0.4 V).

One important factor leading to capacity decay is polysulfide/selenide shuttle behavior between cathode and anode during deep charge–discharge cycles.^[14,31,28] Here in order to verify the factors contributing to the gradual capacity fade in Se/C cathode and S/C cathode, X-ray photoelectron spectroscopy (XPS) of cycled Mg electrode have been investigated as shown in Figure S12 (Supporting Information). Obviously, there exists the selenium and sulfur deposition on the surface of Mg anodes. The cycled Mg foils coupled with both Se/C cathode and S/C cathode darken dramatically, suggesting the deterioration of Mg electrodes. These results reveal the fact that polysulfide/selenide shuttling effects are to a large degree responsible for the capacity decay during deep charge–discharge cycles in consistent with the results of previous reports.^[14,31,28]

The amounts of electrolytes also affect the magnesium–selenium and magnesium–sulfur cell's performances, especially the cycling stability. Reducing the amounts of electrolytes may suppress polysulfide/selenide disproportionation, while it can also increase the viscosity of the polysulfide/selenide solution. The optimized electrolyte amounts lead to a maximum electrode

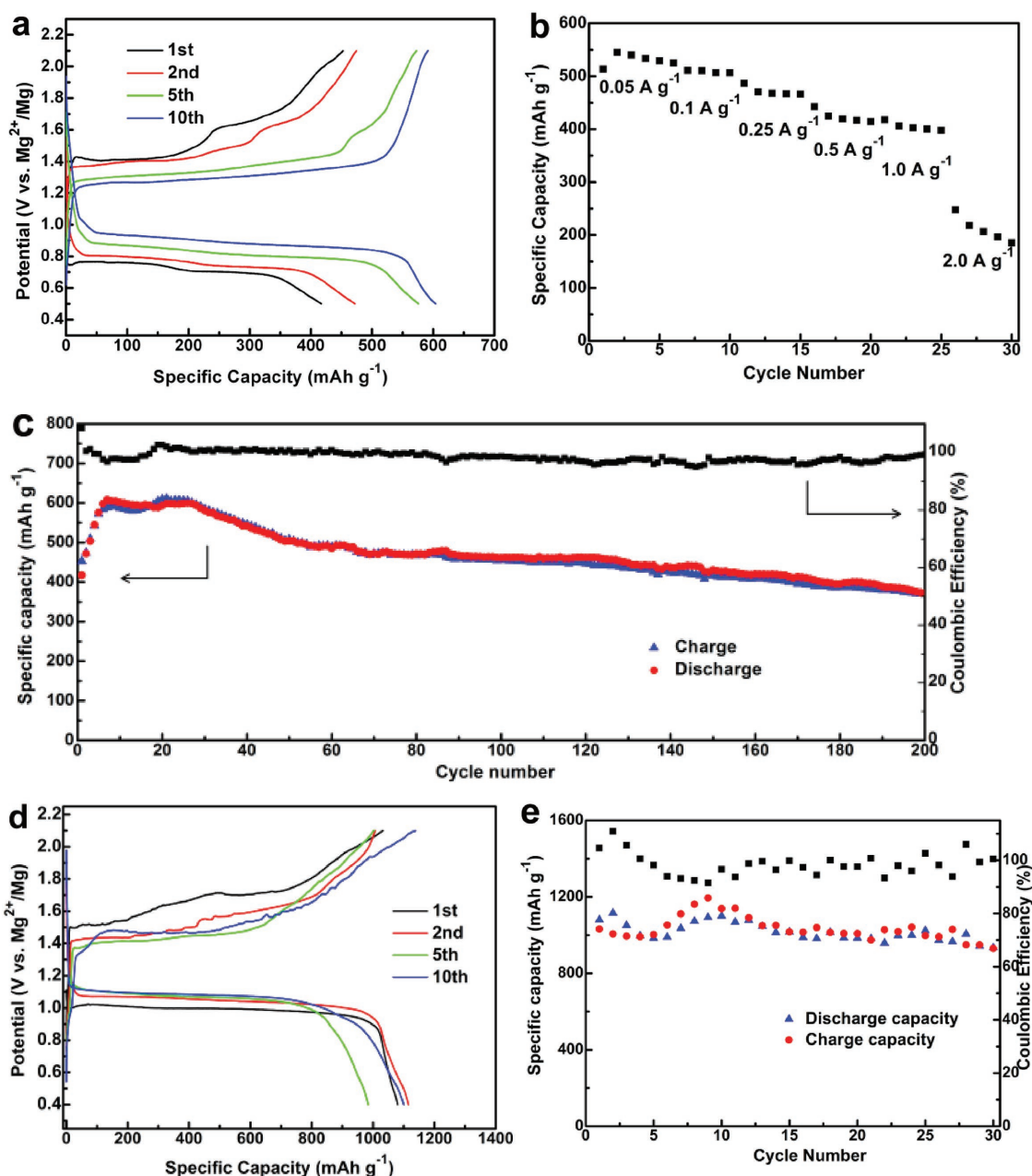


Figure 6. The electrochemical performances of the Se/C and S/C electrodes in BCM electrolyte. a) Galvanostatically charge/discharge profile of the Se/C electrode in a coin cell at a current density of 0.066 A g^{-1} , b) rate performance, and c) cycling stability of the Se/C electrode in BCM electrolyte. d) Galvanostatically charge/discharge profile and e) cycling stability of the S/C electrode in a coin cell containing BCM electrolyte at a current density of 0.05 A g^{-1} .

utility and long-term cycling stability. Figure S13 (Supporting Information) shows the galvanostatically charge/discharge profile of the Se/C electrode in a coin cell containing different amounts of BCM electrolytes. It is worthy noted that the Se/C electrode displays highest specific capacity using $120 \mu\text{L}$ BCM electrolytes. In this work, our aim is to introduce new knowledge to design high-performance Mg-compatible electrolytes. Considering the complexity of our novel magnesium–selenium and magnesium–sulfur systems, detailed and comprehensive explanations on these aspects should be further investigated.

3. Conclusions

In summary, we propose three key design strategies toward exploring efficient B-centered anions for all-round practical Mg salts. The Mg-compatible BCM electrolyte is successfully presented by facile one-step mixing of THFPB and MgF_2 in DME. Incipient species are determined as $[\text{FTHB}]^-$ and $[\text{Mg}(\text{DME})_n]^{2+}$ for Mg stripping/plating processes by MS, NMR, and Raman investigations. During the extensive CV cycles, a new fluorinated borate anion ($[\text{TrHB}]^-$) might be generated. The

Table 2. Comparison of electrochemical properties relating to magnesium–sulfur batteries with previous reports. ($x = 0.065 \text{ mA cm}^{-1}$).

Cathodes	Sulfur loading [%]	Electrolyte	Initial capacity/ $\text{mA h g}^{-1} [\text{A g}^{-1}]$	Capacity retention % [cycle]	Overpotentials [V]
S–C ^[14]	61	HMDS–Mg	1200 (50 μA)	32.8% (2)	≈ 1.11
S–C ^[28]	62	HMDS–Mg	800 (0.02)	32.5% (20)	≈ 0.8
S–rGO ^[34]	49	HMDS–Mg	1024 (0.02)	21.4% (50)	> 1.0
CMK3–S ^[35]	70	Mg(TFSI) ₂	250 (x)	96% (4)	> 1.0
ACC/S ^[31]	15	Mg–HMDS/LiTFSI	390 (0.071)	1000 mA h g^{-1} (30)	≈ 1.0
NG–S ^[33]	50	[Mg(THF) ₆][AlCl ₄] ₂	700 (0.017)	10% (20)	> 1.5
S–CNF ^[36]	50	HMDS–Mg	≈ 1150 (0.033)	$\approx 65\%$ (20)	≈ 0.7
Our work	85	BCM electrolyte	1081 (0.05)	86.4% (30)	≈ 0.4

optimized BCM electrolyte displays 99.8% CEs, 1.1 mS cm^{-1} ionic conductivity, and an enlarged voltage window of Mg-ion electrolytes up to 3.5 V versus Mg within typical coin cells. More surprisingly, the BCM electrolyte supports the conversion reactions of Se/C cathode, achieving highly reversible capacity of 450–600 mA h g^{-1} for more than 100 cycles. As well the Se/C cathode delivers unprecedented high rate capability (405 mA h g^{-1} at a practical rate of 1.0 A g^{-1}). The magnesium–sulfur battery platform shows high average discharge capacity up to 1081 mA h g^{-1} among first 30 cycles with one flat voltage plateau of $\approx 1.1 \text{ V}$ versus Mg, resulting in a high energy density beyond 900 W h kg^{-1} based on the weight of the active electrode materials. This promising BCM electrolyte offers the most possibility as a model electrolyte for future high-energy Mg batteries by examining high-voltage (such as MnO_2) and high-capacity (such as sulfur) cathodes. As well, the novel design concepts could shed light on the exploration directions toward high-voltage Mg-compatible electrolytes. We believe that these significant breakthroughs can highly push forward the developments of practical rechargeable Mg batteries.

4. Experimental Section

Synthesis of BCM Electrolyte: The synthetic procedures of all electrolytes were carried out within an Ar-filled glovebox (Mikrouna) with moisture and oxygen concentrations less than 0.1 ppm at room temperature. The BCM electrolyte was prepared by facile dissolution of both THFPB and MgF_2 salts into DME solutions. In a typical procedure, 1.024 g THFPB was dissolved in 2 mL anhydrous DME followed by adding 0.0062 g MgF_2 . The above electrolyte was stirred for ten minutes and rested for one night prior to use. In order to investigate the effect of solvents on the Mg stripping/plating process, various solvents including THF, DMSO, propylene carbonate, acetonitrile (CH_3CN), diglyme (G2), triglyme (G3), and polyethylene glycol dimethyl ether were used to replace DME to synthesize the BCM electrolytes while keeping the concentration of THFPB and MgF_2 unchanged. To access different borates/ MgF_2 ratios, the concentration of THFPB was varied from 0 to 0.05, 0.1, 0.2, 0.5, and 1.0 M by adding different amounts of THFPB in the BCM electrolytes while keeping the weight of DME solvent and MgF_2 unchanged.

The synthetic procedures of other electrolytes including MgBSB electrolyte, MgDFOB electrolyte, all phenyl complex (APC) electrolyte, and HMDS–Mg-based electrolytes are depicted in the Supporting Information.

The preparations of Se/C and S/C composites are presented in the Supporting Information. As well, the electrode fabrication process,

detailed electrochemical investigations (such as CV, LSV, electrochemical impedance spectroscopy (EIS), and galvanostatically charge/discharge tests), and materials characterizations (including SEM, XRD, Raman spectra, XPS spectra, NMR, electrospray ionization-mass spectrometry (ESI-MS) as well as density functional theory (DFT) calculations) can also be identified in the Supporting Information.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Z.Z. and Z.C. contributed equally to this work. G.C. and Z.C. conceived the idea of the BCM electrolytes with the help of Z.Z.. G.C. and Z.Z. conceived the idea of the magnesium–selenium and magnesium–sulfur batteries with the help of Z.C.. Z.Z. X.W. and S.L. synthesized the Se/C and S/C composites. Z.Z., Z.C., L.Q. and H.X. executed the electrolyte preparation, materials characterization, and electrochemical measurements. Z.Z., Z.C., L.Q., H.X., P.H., X.Z., S.D., Z.L., and G.C. contributed to results analysis and discussion. Z.Z. wrote the first draft with the help of Z.C.; G.C. and L.C. coordinated it. This work was supported by the National High Technology Research and Development Program of China (863 Program, Grant No. 2014AA052303), the National Natural Science Foundation of China program (Grant No. 21271180), and the Key Technology Research Projects of Qingdao (Grant No.13-CX-10).

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